Scalability & Future Plan

Date	28 September 2026
Team ID	6ab221d587602313e82c60c9
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Import is run manually using Load Data	Data must be uploaded by a user each time, which takes effort and can be missed	Medium
2	Only one spreadsheet column structure is supported	Field mapping breaks if the column names or order are changed	Medium
3	Limited error notification for invalid rows	Invalid rows are only ignored or logged, so failures may go unnoticed	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	One admin user runs the import manually	Give a dedicated import role so multiple authorized users can run imports
2	Data Storage	Import Set and Employee Test tables hold small sample data	Clean up old Import Set rows regularly and archive old records
3	Performance	Small files (a few rows) transform in about 1 second	Test with larger files, optimize transform scripts and run imports off-peak using a Scheduled Import
4	Security	Access is with the admin role on a developer instance	Add ACLs and role-based access to control who can load and transform data

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Spreadsheet import using Import Set, Transform Map and Coalesce (completed)	Completed	Working base import process without duplicates
Phase 2	Scheduled Import for automatic loading of spreadsheet data	1 month	Less manual effort and regular data updates
Phase 3	Email notification for failed rows and stronger validation scripts	2 months	Faster identification and fixing of data errors
Phase 4	Support for multiple spreadsheet templates, larger data volumes and role-based ACLs	3 months	Handles more users and bigger datasets securely